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Measuring Housing Stress: How Much do Definitions Matter?

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ABSTRACT *There has been extensive debate in Australia in recent years about the most appropriate measure of housing stress. This article presents estimates of housing stress in Australia and examines the comparative effects of various definitions on the proportion of households in housing stress. Six measures of housing stress are tested, all with different types of income and criteria. Up-to-date estimates of housing stress at the national and state/territory level are produced from the 2005–2006 Survey of Income and Housing from the Australian Bureau of Statistics. These estimates are then compared to identify the least sensitive indicator of housing stress across household tenure type.*

内容提要：近年来，澳大利亚对何为测量住房压力的最佳方法争论颇多。本文论述对澳大利亚住房压力水平的各种估算，考察对家户在住房压力中所起作用的各种说法，各自导致了什么结果。文章检测了对住房压力的 6 中测量方法，这 6 种方法使用了各不相同的收入类型和标准。迄今为止，全国和各省的住房压力估算使用的都是澳大利亚统计局“2005-06 年收入和住房调查”中的数据。文章对这些估算进行比较，试图发现各种住宅所有权类型中对住房压力最不敏感的指征。

KEY WORDS: Housing stress, housing affordability, ratio measures, Australia

Introduction

Decent housing is essential for maintaining an acceptable standard of living for families. It provides warmth, security, safety and privacy to all members of the family. It also provides facilities for cooking, sleeping and entertainment. In Australia, we would expect it to be affordable to people from all socio-economic strata.

Housing has increasingly been recognised as a right of individuals (King, 2003) and any democratic government is expected to work towards protecting and realising this right.

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Erroneous public policies that affect the production and distribution of housing can result in certain groups being socially excluded as they are denied control over their daily lives, or their level of civic participation is weakened (Somerville, 1998). Low-income households and young families are often found to be struggling to meet housing expenses and are exposed to social exclusion in an unaffordable housing market (Healy & Hillman, 2008; Yates, 2008). It is therefore understandable that decent and affordable housing has been seen as an important tool to tackle social exclusion or to maintain social cohesion (Berry, 2003; Frazer, 2005).

To apply this tool, however, one needs to be able to identify the groups experiencing housing affordability problems. A simple and widely used approach to identify housing affordability issues is to employ financial indicators to identify individuals and households in housing stress. This article focuses on the measurement of housing stress—the financial aspects of the housing problem—in Australia.

Housing stress has been recognised as an important economic challenge for families and a salient public policy challenge for governments in Australia (National Housing Strategy, 1992; Tanton *et al.*, 2008). Recently, this was reflected in a 2008 Senate inquiry into housing affordability and in Prime Ministerial level concerns regarding housing affordability problems in Australia (Commonwealth of Australia, 2008). However, there is no general consensus about the method of measuring this important public policy concept. A typical method of measuring housing stress defines a household or a family to be in housing stress if it spends more than 30 per cent of its *income* on housing costs. But is the most appropriate measure here disposable (after-tax) income or gross income? What are the implications of limiting the definition of housing stress to households in the bottom 40 per cent of the income distribution? How are the results affected if we exclude the 10 per cent of the population with the lowest household incomes?

These may sound like obscure technical issues of measurement. Yet such measurement issues have become a matter of serious debate in Australia, not only among academic researchers but also among policy makers. For example, in April 2008, the Australian Reserve Bank Governor expressed the view that the popular measure of housing stress (which is only based on the 30 per cent rule without taking account of the income distribution) is misleading (*The Australian*, 2008). He was of the opinion that the measure should have been limited to those in the bottom 40 per cent of income earners. The Senate Select Committee report listed a number of groups advocating such a '30/40' rule, including the Department of Families, Housing, Community Services and Indigenous Affairs; the Australian Council of Social Services; and a number of prominent academics engaged in housing research (Commonwealth of Australia, 2008, p. 37).

In a further recent development, the Australian Social Inclusion Board used a different measure of housing stress as a measure of social exclusion (Australian Social Inclusion Board, 2010). The indicator used in this report was *lower income* (those earning more than 10 per cent but less than 40 per cent of the income distribution) *private renters* with housing costs exceeding 30 per cent of their gross household income, or what we call in this study, the '30/10–40 rule' applied to renters only (not owners or purchasers). This is a departure from the 30/40 rule used by most government departments and academics in Australia, and clearly suggests that there is a need for a detailed analysis of the implications of choosing one or the other measures of housing stress. This article aims to address this need, and while it uses data for Australian states and territories, the conclusions are applicable to countries using similar measures of housing stress.

This article is structured as follows. First, common variants of the ratio measures used in previous Australian studies are reviewed. Next, data and methods used in this study are described. The next section compares and contrasts the estimates of housing stress corresponding to the various definitions examined in this study. Finally, the article concludes by discussing the findings of the study.

Measures in Previous Australian Studies

Internationally, various measures of housing stress have been used to show how hard it is for a household to meet their housing costs (Whitehead, 1991; Hulchanski, 1995; Stone, 2006; Yates & Gabriel, 2006; Yuen *et al.*, 2006; Paris, 2007; Rea *et al.*, 2008). Despite some criticisms, the ratio approach remains the most widely recognised measure of housing stress (Stone, 2006). A typical ratio measure is based on the ratio of housing costs to income, with the 30 per cent level being a benchmark (Rea *et al.*, 2008).

There are a number of variations on this ratio measure. As Hulchanski (1995) notes, there has been a shift in defining the housing cost-to-income threshold, from 20 per cent in the 1950s to 25 per cent until the 1980s and to 30 per cent now. Most contemporary analyses define households as being in housing stress if they are paying more than 30 per cent of their income on housing costs.

This 30 per cent rule is also used by many home loan lenders in Australia as a guide to how much a household can borrow—and is used by real estate agents as a guide to whether a lessee can afford to rent a property. The rationale behind it is that if a household is paying more than 30 per cent of its income on housing costs, there is not as much income for other essential goods and services such as food, energy, transport, education and health.

A number of different definitions and criteria have been used in estimating housing stress in Australia. The most common method appears to be one that defines a household or a family to be in housing stress if it spends more than 30 per cent of its income on housing costs and also belongs to the bottom 40 per cent of the income distribution. This approach is referred to in short as the '30/40' rule.

This definition was used in the early 1990s by the Australian National Housing Strategy (1992, p. 13) to estimate income units (an income unit is made up of a household member or members who have assumed shared command over an income) in housing stress in Australia. This is based on the assumption that those whose incomes place them above the 40 per cent cut-off should be able to meet market rates of housing costs within the limits of 30 per cent of spending. At income levels above 40 per cent of the income distribution, it may be that these households or families choose to spend more than 30 per cent of their income on housing to live in a higher standard dwelling, or a dwelling closer to the city or work, or to speed up the mortgage repayment.

Some studies have applied this definition to estimate the number of Australian families in housing stress (Harding *et al.*, 2004), while others have used it to calculate households in housing stress (ABS, 2002). The most recent application of this rule was in figures used by the Australian Prime Minister and subsequently published by the Australian Government Department of Families, Housing, Community Services and Indigenous Affairs (FaHCSIA, 2008).

Another common variant of housing stress measurement defines a household to be in housing stress if it spends more than 30 per cent of its income on housing. That is, the relative income (so the income compared to all other Australians) of the household is *not*

taken into account in this definition, only the absolute income is considered when assessing the 30 per cent rule. This method is known as the '30-only' rule. The most recent applications of this method can be found in Marks and Sedgwick (2008) and Tanton *et al.*, (2008), and these two articles show different estimates of housing stress when using gross and disposable income. While Marks and Sedgwick used gross income as the base for constructing housing stress indicators, Tanton *et al.* used disposable income. Analysing longitudinal panel data from the Melbourne Institute's Household, Income and Labour Dynamics in Australia Survey (HILDA), Marks and Sedgwick, using disposable income, estimated 12 per cent of Australian households being in housing stress in 2006. Using the Australian Bureau of Statistics (ABS) Survey of Income and Housing 2005–2006, Tanton *et al.*, using gross income, found that 22.5 per cent of Australian households were in housing stress in 2005–2006.

While some of this difference is going to come from different data-sets and samples, an interesting question is how much of it comes from using gross rather than disposable income when using a 30-only rule. In fact, this article shows that using gross income and a 30-only rule, the 2005–2006 Survey of Income and Housing gives comparable estimates to the HILDA (about 16 per cent). The remaining difference between these two estimates could be attributed to different data-sets, sampling size and sampling methods. One difference is that the HILDA is a longitudinal survey, while the Survey of Income and Housing is designed as a cross-sectional survey. Another difference is the Survey of Income and Housing sample is much larger, and is changed each survey; whereas the HILDA sample, due to the longitudinal nature of the survey, is relatively constant over time.

A less common variant, which has been used by the ABS in the past, limits the definition of housing stress to those households that are paying more than 30 per cent of their income on housing *and* belong to the bottom 10–40 per cent of the income distribution. This method can be referred to as the '30/10–40' rule. The reason for dropping the bottom 10 per cent of the income distribution is due to the concern that the reported incomes of the households in the bottom 10 per cent of the income distribution do not always accurately reflect their living standards and their inclusion in the definition may overestimate housing stress (ABS, 2005).

As an example, in a 2005 report titled *Housing Occupancy and Costs, Australia, 2002–03*, the ABS (2005) applied the '30/10–40' rule to calculate level and trends in housing stress. Using gross income as the base, it was estimated that 5.5 per cent of households were in housing stress in 2002–2003. Although the housing cost to income ratio was based on gross income, the 40 per cent income rule was based on equivalised disposable income. In a previous publication, *Measuring Australia's Progress*, the ABS (2002) used the '30/40' variant and calculated housing stress on the basis of the proportion of disposable income spent to meet housing costs. Using this method, housing stress was estimated to be around 11 per cent between 1995 and 1998 (ABS, 2002). This was almost double the level estimated by applying the '30/10–40' rule to gross income (ABS, 2005).

From the above research, it can be seen that not only are there different definitions in terms of the income cut-off (none, 40 per cent, 10–40 per cent), but that researchers have also used two different bases for the income measure. While some researchers use gross income, others use disposable (after-tax) income to calculate housing stress. The choice seems to be driven largely by the nature of data. For example, the Australian Census only

has data on gross income, and this is in groups rather than individual incomes. Therefore, any estimates based on these data are constrained to using gross income in groups.

Randolph and Holloway (2002) used gross income to examine housing stress among low-income households in Sydney (income below the median household income in New South Wales). They used the 1996 Census data and the 1999 Australian Housing Survey and estimated that 47 per cent of low-income households in Sydney were in housing stress in 1999. Further, as cited previously in this section, ABS (2005) tends to use gross income as the base for identifying those paying more than 30 per cent of their income on housing, but then disposable income for determining the 40 per cent income cut-off.

It can be seen from this review of the literature that there are many different options in calculating housing stress, which result in many different estimates of the proportion of people in housing stress. In this article, we compare six major options and examine their impact on housing stress rates using national and state estimates. The next section defines the variants of housing stress examined in this article, lists the data sources and describes the methodology applied to estimate these indicators.

Methodology

Definitions

This study examines three definitions of housing stress.

(1) '30-only rule'. This is the most basic of the ratio measures of housing stress. According to this rule, a household is defined to be in housing stress if it spends more than 30 per cent of its disposable or gross income on housing costs.
(2) '30/40 rule'. This is a narrower variant of the '30-only rule'. In this rule, a household is said to be in housing stress if it spends more than 30 per cent of its disposable or gross income on housing costs and the household also belongs to the bottom 40 per cent of the equivalised disposable income distribution.
(3) '30/10–40 rule'. This approach goes one step further to exclude the bottom 10 per cent of the income distribution. A household is defined as being in housing stress if it spends more than 30 per cent of its disposable or gross income on housing and falls in the bottom 10–40 per cent of the equivalised disposable income distribution.

In all the definitions, households with negative and nil incomes have been removed from the analysis. A small number of households have reported nil or negative incomes in the ABS surveys. These are often excluded from any analysis related to income distribution and financial well-being, as research from the ABS (2005) has shown that the expenditure of many of these households is similar to that of households earning much more, so these incomes are considered an unreliable guide to a household's standard of living.

Two types of income, gross and disposable, are considered as the base to calculate the housing cost to income ratio (the 30 per cent rule). Gross income is the income from all sources before deducting tax and the Australian Government's medical insurance (Medicare) levy. Disposable income is the income that remains to a household after deducting estimated personal income tax and the Medicare levy from gross income.

Researchers use 'equivalised' income measures to take account of the number of adults and children that each household's income has to support, so as to put households of

differing composition on a more equal footing when assessing who is well-off and who is not. To calculate equivalised disposable income, which is used for the income distribution criteria (the 40 and 10–40 rules), the income of a household is first equivalised using the modified-OECD equivalence scale, which assigns a weight of 1 point to the first adult, 0.5 to each of the remaining adults aged 15 years or over, and 0.3 points to each of the dependent children under 15 years in the household. Using this factor, the household income is standardised, so all households have an income expressed relative to that of a single person household.

Once equivalised disposable household income has been calculated, all households are then ranked by these incomes from the lowest to the highest, and the people in a household are then put into 10 equal groups (deciles) in order to identify people with household incomes in the bottom 40 per cent of the income distribution. Thus each income decile is a 'person-weighted' decile and we have an equal number of people (not households) in each decile.

Housing costs are defined as the regular expenses of a household in providing for their accommodation. The data on housing costs refer to the payments related to the respondent's usual place of residence (ABS, 2007b, p. 22). Housing costs include rents for renters, general and water rates payments for owners—and, in the case of owners with mortgages, includes mortgage repayments for secured or unsecured loans taken primarily for housing purposes (including, for example, to build, or to alter or add to the current dwelling). It does not include repayments on loans for cars, overseas holidays and so on, which are collected separately in the survey.

Data

The data used for the national and state/territory level estimates are from the 2005–2006 Survey of Income and Housing (SIH). At the time of undertaking this research, this was the latest income survey available from the ABS. This data-set contains information on 9961 households, comprising 51 households with negative or nil income and 9910 households with positive income. The direct national level estimates of housing stress presented in the next section are based on the 9910 households. Some of our results are produced at a state/territory level, and it is important to note that the Confidentialised Unit Record File of the SIH combines the two Australian territories, the Australian Capital Territory and the Northern Territory, to retain data confidentiality. As these two territories have very different demographic and income profiles, it is very difficult to interpret results produced from the combined SIH data. Thus, our estimates of housing stress in the Australian Capital Territory/Northern Territory should be used only to examine the sensitivity of these results to different measures of housing stress, not to draw conclusions about actual rates of housing stress in either territory.

Results

Figure 1 shows that, at the Australian national and state level, housing stress prevalence rates are lower when calculated with gross income than disposable income. In 2005–2006, just over 22 per cent of households were spending more than 30 per cent of their disposable income on housing. This figure dropped by about one-fourth, to 17 per cent, when the calculation was based on gross income. For the other two definitions, there were

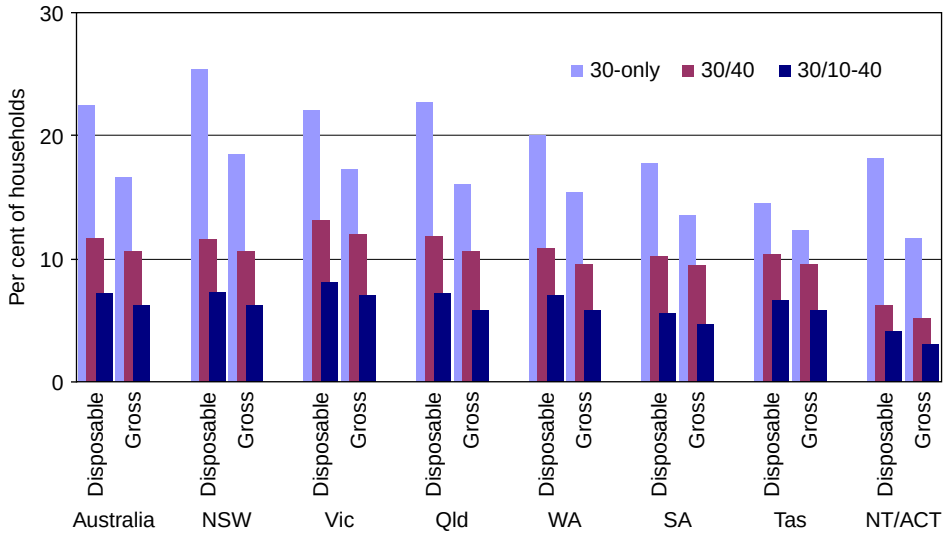


Figure 1. Percentage of households in housing stress according to various definitions, by states or territories, Australia, 2005–2006. *Note:* NSW = New South Wales, Vic. = Victoria, Qld = Queensland, WA = Western Australia, SA = South Australia, Tas. = Tasmania, NT = Northern Territory, ACT = Australian Capital Territory. 'Disposable' refers to disposable income while 'Gross' refers to gross income being used to categorise which households have lower incomes. *Source:* 2005–2006 Survey of Income and Housing confidentialised unit record file; authors' calculations.

smaller gaps in prevalence rates arising from the type of income used. For example, under the '30/40' rule, the housing stress rate stands at about 12 per cent if we choose disposable income as the base, and at about 11 per cent if it is based on gross income. If we go further to exclude the poorest 10 per cent of households ('30/10–40' rule), housing stress is estimated at about 7 per cent with disposable income and about 6 per cent with gross income.

At the national level, the gap in the estimates of housing stress arising from choice of income is the narrowest under the '30/40' rule. This pattern holds true across all the states and territories.

Figure 1 also shows that the estimated proportion of households in housing stress drops substantially as we narrow down the definition. Nationally, with disposable income as the base, the move from the 30-only rule to the 30/40 rule means the housing stress rate drops by nearly half, from above 22 per cent to less than 12 per cent. It drops again by more than a third, to just above 7 per cent, when we exclude the lowest income decile. In the case of gross-income-based measures, the magnitude of the decline is slightly sharper going from the 30/40 to the 30/10–40 rule than for the 30-only to the 30/40 rule.

Another important insight from these results, as summarised in Table 1, is that the ranking of states/territories in terms of housing stress rates varies with the measure used. Under the basic 30-only rule and disposable income, the housing stress rate is the highest in New South Wales followed by Queensland and Victoria and the lowest in Tasmania. But with gross income, though New South Wales remains the state with the highest rates of housing stress, Victoria climbs to the second highest rank, pushing Queensland down to third position; and the Northern Territory and Australian Capital Territory are placed in

Table 1. Ranking (1–7) of states/territories from the highest to the lowest rates of housing stress under various definitions, 2005–2006

	Disposable income			Gross income		
	30-only	30/40	30/10–40	30-only	30/40	30/10–40
NSW	1	3	2	1	2	2
Qld	2	2	3	3	2	3
Vic.	3	1	1	2	1	1
WA	4	4	4	4	4	3
NT/ACT	5	7	7	7	7	7
SA	6	6	6	5	6	6
Tas.	7	5	5	6	5	5

Source: ABS Survey of Income and Housing Costs 2005–2006; authors' calculations based on housing stress rates. Same rank is assigned for states/territories that have equal rates.

the bottom as Tasmania climbs to the second lowest position. So the effect on the state rankings of using gross income is large when the 30-only rule is used.

Looking at the state rankings using the 30/40 rule, it can be seen that the rankings don't change as much going from 30/40 disposable to 30/40 gross. Only NSW and Queensland have changed slightly. The same is true of the 30/10–40 rule. The rank for Western Australia has changed slightly, but otherwise there was no large change. This is one of the main conclusions from this analysis; the choice of gross and disposable income has little effect on housing stress for the 30/40 and 30/10–40 rule, but it does for the 30-only rule.

Figure 2 presents housing stress prevalence rates across four tenure types: owners, purchasers, private renters and public renters. In these data, the term 'owners' refers to outright owners of the dwelling occupied at the time of the survey, and the term 'purchasers'

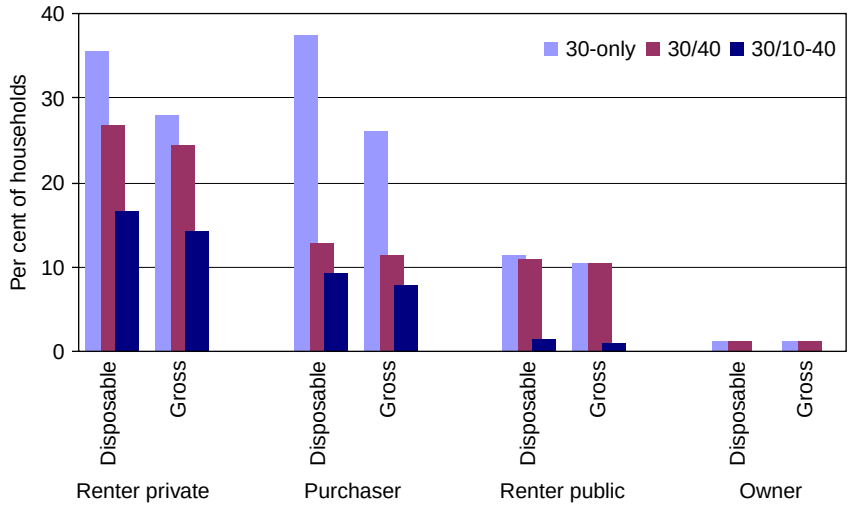


Figure 2. Percentage of households in housing stress according to various definitions, by tenure type, Australia, 2005–2006. *Note:* 'Owner' refers to outright home owner. 'Purchaser' refers to owner with mortgage. *Source:* 2005–2006 Survey of Income and Housing confidentialised unit record file, authors' calculations.

refers to owners with a mortgage. Households paying rent to a real estate agent or to another person not in the same household are defined as private renters and those paying rent to a state or territory housing authority or trust are classified as public renters (ABS, 2007a).

As shown in Figure 2, housing stress among owners is low, around 1 per cent, irrespective of whether it is based on disposable or gross income. It is worth noting that the housing stress rate approaches nil for outright home owners when the poorest 10 per cent of the households are excluded from the definition.

Similarly, the prevalence of housing stress drops to a very low level, around 1 per cent, among public renters if the poorest 10 per cent of households are excluded. Otherwise, as many as 1 in 10 renters in public housing can be defined to be in housing stress under either of the other two rules. This does suggest that the 30/10–40 rule is excluding low-income households, such as public housing tenants, that should be counted as being in housing stress. However, the housing stress rates for public housing tenants seems to be inconsistent with the fact that they are living in subsidised housing, and in Australia, are required to pay only 25 per cent of their income on housing costs. This issue will be looked at further in the Discussion section.

Among home purchasers (owners with mortgages), under the 30-only rule, substantially different housing stress rates are obtained using disposable income (38 per cent) compared to gross income (26 per cent). Both figures drop remarkably, by more than one-half to two-thirds, when the measurement is confined to the bottom 40 per cent of the income distribution. This may be seen as suggesting that many of these purchasers, who appear in housing stress when the full income distribution is used, are higher income households voluntarily paying more into their housing loan—or perhaps they may be choosing to live in more expensive housing (close to city centres or larger houses). An alternative interpretation might be that such households were not in housing stress when they first took on their mortgages, but that a succession of interest rate increases experienced in Australia, from a low of 6.05 per cent in 2002 to 7.8 per cent in August 2006 have now pushed them into a more vulnerable position.

For private renters, the choice of definition as well as income type makes a large difference. Under the first rule, housing stress is 36 per cent with disposable income as the base and 26 per cent with gross income—a gap of 10 percentage points. This is a large difference, although not as large as the difference for purchasers. Gaps in prevalence rates with disposable and gross incomes were much smaller under the '30/40' rule (27 per cent compared to 24 per cent) as well as under the '30/10–40' rule (17 per cent compared to 14 per cent).

It is interesting to note that the ranking of housing stress by tenure type varies across the different definitions of housing stress. Under the '30-only' rule and disposable income, home purchasers appear to be the most stressed group, slightly ahead of private renters. But once the definition is narrowed by excluding households above the second quintile of the income distribution, private renters stand out as the most stressed group, well above the purchasers. The exclusion of the poorest 10 per cent of households does not change the relative rankings, though it drastically diminishes the housing stress rate among public renters.

Overall, the choice of the definition of housing stress (income cut-off) has a greater impact on the estimates of housing stress than the type of income (disposable vs. gross) used as the base measure of resources.

Discussion

This study was inspired by the observation that there is no common practice with regard to choice of income type and income thresholds in measuring housing stress in Australia. Some studies have used gross income while others have preferred disposable income. There are varying practices with regards to setting income thresholds. Some studies look at all households irrespective of their income level, some tend to consider only the households that are in the bottom 40 per cent of income distribution, and others argue for excluding the households in the bottom 10 per cent of the income distribution.

We have found that there is not much difference between using gross and disposable income when we apply the 30/40 or the 30/10–40 definition of housing stress; however, there is a notable difference when applying the 30-only definition. This is an interesting result, as it suggests that the '30-only' definition is more susceptible to the choice of type of income than are the other definitions. This may be due to larger differences between disposable and gross income at the higher rather than lower end of the income distribution, owing to the progressive nature of the Australian tax system.

The other issue we found with the '30-only' definition was that it showed a very high proportion of purchasers in housing stress. Like some other commentators (for example, Yates, 2008), we think this may be due to some very high-income purchasers paying more than 30 per cent of their income on the loan voluntarily, by either choosing a more expensive house, knowing they can afford higher repayments; or paying more to pay off the loan earlier. It is possible for them to commit a higher ratio, say 40 per cent, of their income to a mortgage repayment as the remaining 60 per cent is still much larger than the remaining 70 per cent of a low-income household (Paris, 2007). Alternatively, it may be due to some purchaser households being pushed beyond the 30 per cent threshold by the increases in interest rates experienced in Australia from 2002 to 2006. When the low-income criterion is added, the proportion of purchasers in housing stress drops from 35 per cent (using disposable income) to about 12 per cent. This is a considerably wide gap that may put policy makers in a dilemma if they have to formulate a financial assistance package.

We also found that the 30/10–40 definition excluded many public housing tenants from housing stress. This was a major concern for us, and the national housing stress rate for public housing tenants dropped from nearly 10 per cent using the 30/40 definition to 1 per cent using the 30/10–40 definition. This was a significant decline because the 30/10–40 definition excluded a large proportion of public renters who experience very low incomes (in the bottom 10 per cent of the income distribution).

These public renters qualify for public housing in Australia because they are low-income families. The amount they pay for rent is typically calculated as 25 per cent of their assessable income, which is more like gross income but excludes specific purpose payments (McNelis, 2006). So in theory there should be no public housing tenants in housing stress. This then raises a question about whether public renters in the survey might have under-estimated their income or over-estimated their rents. Administrative data compiled by the Australian Institute of Health and Welfare (AIHW, 2006, p. 21) show that 99.7 per cent of rebated public housing tenants spent not more than 30 per cent of their income in rent in 2005–2006. Yet it is important to note that the rental system in Australia is complex owing to variations across the jurisdictions in the definitions of assessable income, eligibility criteria for rebate and circumstances of renters (McNelis, 2006).

Obviously, the discrepancy in income and housing cost information from administrative and survey data in Australia needs to be investigated in detail in future studies.

Using the 30/10–40 rule also excluded many private renters from being in housing stress. There is no evidence that this is a data quality issue, as was possible with the public renters. The problem with the 30/10–40 rule is that it may exclude renters and owners who have very low incomes and are in housing stress. Research by Saunders and Bradbury (2006) suggest that excluding the bottom 10 per cent of the income distribution in calculating poverty rates is too extreme, and our current research supports this conclusion for calculating rates of housing stress.

Overall, the definition we found most suitable for measuring housing stress was the 30/40 rule. This was only slightly affected by the use of disposable or gross income—and did not produce extraordinarily high or low housing stress rates for any of the sub-groups we looked at. We found the 30-only rule produced very high housing stress rates for purchasers, while the 30/10–40 rule produced very low rates for public housing tenants (although this may also have been due to the quality of the survey data used) and private renters. As such, the 30/40 rule could be more robust than others for a trend analysis of housing stress during a period of significant fluctuations in interest rates.

Further, there are solid conceptual reasons for using the 30/40 rule. In terms of using gross or disposable income, this study has found that there is little difference between housing stress rates calculated by using the 30/40 rule using either measure. If the 30-only rule was being used, then more thought would need to be given to whether to use gross or disposable income, as this choice of income type has a much greater effect on housing stress rates using the 30-only rule. While most poverty researchers would always prefer a measure based on disposable household income, where it is available, because it represents the amount left to a household to spend on housing and other needs, in many cases disposable income is not available (for instance, the Australian Census data only has gross income available). The ABS (2007b, p. 3) also makes the point that "disposable income better represents the economic resources available to meet the needs of households".

With regard to possible future directions for research, the measure used in this analysis is constructed by setting the ratio of income to housing cost at 30 per cent. This 'ratio' approach is a typical way of looking at housing affordability but the threshold has been criticised as being arbitrary (Yates & Gabriel, 2006, p. 32). While there is no compelling reason to set this benchmark at 30 per cent, it has become a standard yardstick for financial institutions and industries. In Australia, this benchmark is used by banks in looking at clients' ability to pay off a loan; and it is also used by rental agencies to determine if a renter can afford a rental property. Thus, while there is some justification for using it in an Australian context, it is more a guideline than a firm rule. Therefore, it is likely to continue to be used unless a convincing alternative measure is developed.

The same threshold has been used for all groups. Future work could look at whether it is necessary to set different benchmarks for different tenure types. This may be an issue for areas that belong to remarkably different geographic locations, such as inner metropolitan areas and sparsely settled regional areas.

This study does not take into account the expenses of transportation, nor does it consider the standard of the dwellings in which households are located, or issues of overcrowding. These may be important issues. For households in peripheral suburbs, housing costs may be slightly lower than in city centres, but a substantial amount of household income may

be expended in commuting to work. Future research may look into the possibilities of adjusting housing cost by taking account of the transportation cost.

Finally, although this study examined the robustness of major variants of the ratio indicator using Australian data, the findings are expected to contribute to the international debate on the choice of housing affordability measure. The international relevance of this study can be highlighted by the fact that the ratio measure based on a certain cost-to-income threshold continues to be used to assess housing affordability in many countries including the USA, the UK, Canada and New Zealand, and that there is a degree of disagreement regarding the choice of the indicators (Kutty, 2005; Robinson *et al.*, 2006; Stone, 2006; Rea *et al.*, 2008). The findings of this study can help refine or substantiate the ratio measure of housing affordability used in various countries.

The main point made in this article is that the different measures of housing stress, and the different types of income used (gross or disposable) can provide significantly different estimates of the proportion of people in housing stress. This is an important point for policy makers within government. A policy for affordable housing based on a 30-only rule might inadvertently provide affordable housing to richer households which may have capacity to service a larger mortgage repayment. In contrast, a policy based on a 30/10–40 rule may miss poorer households (the bottom 10 per cent of the income distribution) that have the greatest need for affordable housing.

The second important point for policy makers is how different definitions of housing stress can be used to drive a certain agenda. In an era of evidence-based policy, the evidence being used has to have sound theoretical arguments for its use over other possible measures. In this article, arguments have been made for certain measures of housing stress on technical and theoretical grounds, and then the implications of each of the different definitions have been tested. Using a theoretical argument to decide which measure of housing stress to use is important for policy makers, to distance themselves from claims of political bias.

In terms of a recommendation for policy analysts, the authors believe that the 30/10–40 rule excludes many people that may legitimately be in housing stress; and the 30-only rule includes many people who may not be struggling to pay their housing costs. The 30-only rule is also affected substantially by the choice of gross or disposable income, which does not affect the 30/40 rule as much. This means that the 30/40 rule is the best among these rules to use when analysing policy.

Conclusions

This study shows that housing stress indicators can be less sensitive and more comparable if they are based on a 30/40 rule. There is little difference between the use of gross and disposable incomes when housing stress is measured by applying the 30/40 or the 30/10–40 variant. We have shown that the 30-only variant is quite sensitive to the choice of income type because of larger differences between disposable and gross income at the higher rather than lower end of the income distribution. We have also found that the 30/10–40 rule excludes people with extremely low incomes, who may genuinely be in housing stress, and so may under-estimate the number of people in housing stress.

The findings of this study are expected to help contribute to an informed choice of housing stress indicators. This study identifies a robust variant of housing stress, that is, the one based on the '30/40' rule, and this indicator can be chosen to examine the impact of housing policies on housing stress over time and across different settings.

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